

HOANG TUAN TU

AI Engineer · Machine Learning · Computer Vision · NLP

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Thanh Xuan, Hanoi

PROFESSIONAL SUMMARY

AI Engineer with 1+ year of hands-on internship experience building production-grade deep learning systems for medical imaging and LLM-powered content pipelines. Proficient in PyTorch, TensorFlow, FastAPI, and C++. Demonstrated ability to deliver measurable results — ~90% DICE score on medical segmentation and ~90% stylistic accuracy in automated content generation. Actively seeking a full-time AI/ML Engineer role to build scalable, real-world AI systems.

WORK EXPERIENCE

VietRad · AI Engineer

Nov 2024 – Apr 2025

- ▶ **Achieved ~90% DICE score** on medical image segmentation by engineering UNet-based deep learning models with PyTorch and custom data augmentation strategies on a proprietary DICOM dataset.
- ▶ **Architected a DICOM communication layer** in C/C++ that enabled stable, low-latency data exchange between medical imaging hardware and backend services — reducing data transfer errors to near zero.
- ▶ Collaborated with medical professionals to validate model outputs, iterating on training pipelines to meet clinical accuracy thresholds.

AI MAX · AI Engineer Intern

Jun 2024 – Sep 2024

- ▶ **Engineered LLM-based content generation pipelines** (Python, Flask, LLM APIs) achieving ~90% stylistic similarity to target creators for automated TikTok and YouTube content — reducing manual editing time by an estimated 60%.
- ▶ **Built multilingual document processing workflows** supporting PDF, DOCX, and slide formats using fitz (PyMuPDF) and python-docx, enabling cross-language content reuse across 3+ language pairs.
- ▶ Designed and deployed end-to-end NLP systems integrating document parsing, structured generation, and language transformation into a single FastAPI-based service.

EDUCATION

VNU Hanoi University of Science

2021 – 2026

B.Sc. Mathematics and Computer Science

TECHNICAL SKILLS

Machine Learning: Scikit-learn, Pandas, NumPy · Supervised/Unsupervised Learning · Feature Engineering · Model Evaluation

Deep Learning: PyTorch, TensorFlow, Keras · Transfer Learning · Optimization · Model Deployment

Computer Vision: Object Detection (YOLO) · Face Recognition (FaceNet) · Image Segmentation (UNet) · Filtering

NLP / LLM: LLM API Integration · Prompt Engineering · Multilingual Text Processing · Document Parsing

Backend: FastAPI, Flask, Django · REST API Design · C/C++ Backend · PostgreSQL

Tools & DevOps: Git/GitHub · CI/CD · Docker (basic) · Linux

Languages: Python (advanced) · C/C++ (intermediate) · Java (intermediate) · English B2

PERSONAL PROJECTS

Face Attendance System *PyTorch · YOLO · FaceNet · FastAPI · PostgreSQL*

- ▶ Designed real-time face detection and recognition pipeline using YOLO for detection and MobileFaceNet for feature extraction; identification via cosine similarity against a PostgreSQL-backed feature store.
- ▶ github.com/Grizmo2610/Face-Attendance-System

Age Estimation from Biometric Images *PyTorch · Multi-Task CNN · Transfer Learning*

- Developed multi-task deep learning model predicting subject age from images with ~4.0 RMSE; leveraged auxiliary attribute prediction tasks to improve age estimation accuracy.
- github.com/Grizmo2610/AGE-ESTIMATION

Image Colorization via CNN & Federated Learning *PyTorch · CNN · Federated Learning*

- Implemented color propagation (flood fill) image processing using CNN with federated learning architecture to enable privacy-preserving distributed training.
- github.com/Grizmo2610/ColorFill

Anime Recommendation Engine *SVD · Collaborative Filtering · TensorFlow/Keras*

- Processed **71M+ user reviews** to build a collaborative filtering recommendation system achieving **~80%** rating prediction accuracy using SVD and matrix factorization techniques.

AWARDS & RECOGNITION

- **Top 47 — Detect AI vs. Human-Generated Images Competition** · Competitive computer vision challenge (2024)
- **Consolation Prize — Student Scientific Research Conference** · Application of Fourier Transform in signal/image processing